

1. Introduction to Linux

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What is Linux: Linux is a free, open-source Unix-like operating system based on the Linux kernel, created by Linus Torvalds in 1991.

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Linux Distribution: Collections of the Linux kernel plus software (e.g., Ubuntu, CentOS, Fedora, Kali).

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Why Linux?

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Open source

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Stability and security

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Developer-friendly

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Lightweight and customizable

- **Shell and Terminal:**
 - *Shell:* Interface between user and kernel. Common shells: bash, sh, zsh.
 - *Terminal:* Program used to interact with the shell (e.g., GNOME Terminal, Konsole).
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2. The Linux File System

- **Directory Structure (FHS - Filesystem Hierarchy Standard):**
 - /: Root directory
 - /bin: Essential user binaries

- /sbin: System binaries
- /etc: Configuration files
- /dev: Device files
- /proc: Virtual filesystem for system info
- /home: User home directories
- /var: Variable files (logs, mail)
- /tmp: Temporary files
- /boot: Bootloader files
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/mnt, /media: Mount points

- **Inodes:**
 - Metadata for files (permissions, timestamps, size, pointers to data blocks)

3. Working with Files and Directories

- **Commands:**
 - ls, cp, mv, rm, mkdir, rmdir, cd, pwd, touch, find
- **Path Types:**

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Absolute: Starts from /

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Relative: Based on current location

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Hidden Files:

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Start with . (e.g., .bashrc)

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Wildcards:

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*, ?, [] used in file pattern matching

4. Getting Started with Linux

- Login methods: Console, GUI, SSH
- User roles: root vs normal user
- Package Management:
 - Debian-based: apt, dpkg
 - RedHat-based: yum, dnf, rpm

5. Revision of Basic Commands

A recap of the most commonly used Linux commands with use cases:

- ls -l, cp file1 file2, mv, rm -r, cat, more, less, wc, sort, head, tail, etc.
 - man/whatis for help, alias for shortcuts, jobs, ps for process monitoring
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6. Access Control List and chmod

- **Permissions:** Read (r), Write (w), Execute (x)
- **File Mode:** -rwxr-xr--
- **chmod:** Change file permissions
 - Symbolic: chmod u+x file
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Numeric: `chmod 755 file`

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ACL (Access Control List):

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Extended permissions beyond owner/group/others

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`setfacl -m u:username:rwx file`, `getfacl file`

7. **chown Command**

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Changes file ownership

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`chown user file`, `chown user:group file`

8. Network Commands

- telnet: Connects to remote host (port 23, insecure)
 - ftp: File Transfer Protocol client (port 21)
 - ssh: Secure Shell for remote login (port 22)
 - sftp: Secure FTP via SSH
 - finger: User information lookup
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9. Use of Secondary Storage Devices

- Devices: HDD, CDROM, Floppy (legacy)
- Device files: /dev/sda, /dev/cdrom
- Partitioning: fdisk, parted
- Formatting: mkfs.ext4, mkfs.vfat
- Mounting: mount /dev/sdb1 /mnt, umount /mnt
- Checking: df -h, du -sh

10. Lab Practice Commands

Basic and essential commands used for daily administration and scripting:

bash

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ls, cp, mv, lpr, sort, grep, cat, tac, more, head, tail, man, whatis, whereis, locate, find, diff, file, rm, mkdir, rmdir, cd, pwd, ln, ln -s, gzip, gunzip, zip, unzip, tar, zcat, cal, bc, bc -l, banner, date, time, wc, touch, echo, who, finger, w, whoami, who am i, alias, unalias, pushd, popd, jobs, ps